

ROOSEVELT ROADS, PR, USA

WMO: 785350

Lat: 18.255N

Lon: 65.641W

Elev: 33

StdP: 14.68

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 11630

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	68.4	70.1	57.1	69.6	75.6	60.3	78.1	76.6	19.9	78.8	18.5	78.9	2.3	330	0.407

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.4	90.2	79.5	89.4	79.0	88.4	78.4	81.4	86.9	80.8	86.3	80.3	85.8	12.4	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.7	154.4	84.1	79.2	151.8	83.9	78.7	149.2	83.8	44.9	87.2	44.3	86.7	43.7	85.8	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	16.7	15.7	64.0	91.2	1.8	1.2	62.7	92.1	61.6	92.8	60.6	93.4	59.3	94.3		
(5)			61.1	83.1	2.7	1.4	59.1	84.1	57.5	84.9	56.0	85.7	54.1	86.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.7	77.4	77.5	78.2	79.5	81.2	82.7	83.3	83.7	82.9	82.1	80.6	78.6
	DBStd	2.90	1.95	1.90	2.01	2.02	1.94	1.63	1.84	1.64	1.66	1.72	1.90	1.91
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11192	850	769	874	885	967	982	1033	1044	987	996	917	888
	CDD65	5717	385	349	409	435	502	532	568	579	537	531	467	423
	CDH74	56509	2696	2391	3088	3825	5080	6011	6626	6908	6203	5723	4500	3458
	CDH80	16058	338	341	537	818	1318	1924	2344	2609	2245	1896	1148	540
Wind	WSAvg	8.6	8.6	8.8	9.1	9.0	8.7	9.5	9.9	9.2	7.7	6.8	7.4	8.4
Precipitation	PrecAvg	56.8	2.8	2.2	2.5	3.6	6.6	3.8	4.1	5.4	6.0	7.5	6.3	4.7
	PrecMax	95.2	8.3	6.7	5.9	10.5	27.2	9.6	8.5	11.7	16.3	27.4	19.4	16.1
	PrecMin	33.8	0.9	0.9	0.4	0.5	0.5	0.7	1.5	1.8	2.0	2.0	1.3	1.4
	PrecStd	17.0	1.5	1.2	1.5	2.3	5.7	2.2	1.9	2.3	3.3	4.9	4.0	3.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.5	85.9	86.5	88.2	89.1	89.9	90.9	91.0	90.5	90.3	88.4	86.3
		MCWB	73.3	73.0	75.3	75.6	77.0	78.0	80.0	80.2	80.1	79.3	77.5	76.4
	2%	DB	83.9	84.2	84.8	86.4	87.8	88.3	89.9	90.0	89.7	89.0	87.6	84.6
		MCWB	73.6	73.3	74.2	74.7	76.6	77.8	79.1	79.8	79.6	78.6	77.3	75.1
	5%	DB	82.5	83.2	84.0	85.3	86.4	87.7	88.4	88.7	88.4	88.1	86.3	84.0
		MCWB	73.5	73.5	73.8	74.4	76.3	77.6	78.2	79.1	79.2	78.2	77.0	74.9
	10%	DB	81.9	82.1	82.6	84.1	85.5	86.5	87.7	88.0	87.9	87.1	85.2	83.0
		MCWB	73.4	73.1	73.1	74.2	76.2	77.4	77.9	78.7	79.0	77.9	76.6	74.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.0	77.3	78.2	79.8	79.8	80.9	81.6	82.3	82.4	81.6	80.3	79.3
		MCDB	81.3	81.4	82.5	82.5	85.0	85.9	87.8	88.3	88.1	86.6	84.7	84.1
	2%	WB	75.9	75.7	76.4	77.9	78.6	80.1	80.4	81.2	81.4	80.6	79.1	77.6
		MCDB	81.0	81.3	82.2	82.7	84.1	85.5	86.7	87.0	86.7	85.8	84.0	82.2
	5%	WB	75.2	75.0	75.5	76.5	78.1	79.3	79.9	80.4	80.7	80.1	78.3	76.5
		MCDB	80.6	80.8	81.4	82.2	83.7	85.0	86.2	86.3	86.0	85.3	83.5	81.4
	10%	WB	74.6	74.2	74.8	75.8	77.4	78.5	79.1	80.0	80.1	79.3	77.6	75.9
		MCDB	80.0	80.1	80.7	81.7	83.1	84.2	85.4	85.8	85.4	84.6	83.1	80.9
Mean Daily Temperature Range	5% DB	MDBR	9.8	10.1	10.3	10.1	9.4	8.8	9.1	9.4	10.5	10.8	10.2	9.3
		MCDBR	10.2	10.9	11.0	10.9	10.0	9.5	9.9	10.2	11.3	11.7	11.1	10.0
		MCWBR	5.1	5.1	5.1	5.0	4.7	4.4	4.8	5.1	5.4	5.4	5.3	4.9
	5% WB	MCDBR	9.6	10.2	10.3	10.3	9.3	8.9	9.3	9.5	10.6	10.9	9.9	9.1
MCWBR		5.1	5.4	5.2	5.4	4.8	4.7	5.3	5.3	5.6	5.7	5.4	4.9	
Clear-Sky Solar Irradiance	taub	0.376	0.390	0.390	0.415	0.458	0.532	0.533	0.493	0.446	0.397	0.390	0.382	
	taud	2.542	2.494	2.489	2.452	2.338	2.092	2.086	2.228	2.394	2.550	2.562	2.558	
	Ebn at Noon	288	289	292	283	268	246	246	258	272	284	282	283	
	Edh at Noon	31	34	35	37	41	52	52	46	38	32	30	30	
All-Sky Solar Radiation	RadAvg	1481	1727	1921	2018	1945	1966	1997	1947	1835	1622	1431	1379	
	RadStd	46	39	74	108	128	116	102	77	119	113	102	62	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	+0.50	+0.85	N/A	+0.59	N/A	N/A	N/A	N/A	+203	+202

Nomenclature: See separate page